Impact of Refactoring on Test Smells - file level

Refactoring Type	Assertion Roulette	Conditional Test Logic	Constructor Initialization	Default Test	Dependent Test	Duplicate Assert	Eager Test	EmptyTest	Exception Catching Throwing	General Fixture	IgnoredTest	Lazy Test	Magic Number Test	Mystery Guest	NumberOfMethods	Print Statement	Redundant Assertion	Resource Optimism	Sensitive Equality	Sleepy Test	Unknown Test	Verbose Test
Add Conditional Annotation	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Any Refactoring	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Change Assertion Type	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Custom Assertion	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
De-Parameterize Test	0.00	-1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-1.00	0.00	-9.00	0.00	0.00	0.00	0.00	0.00	-1.00	0.00
Delegated Setup	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Grouping Tests	-1.00	0.00	0.00	0.00	0.00	0.00	-1.00	0.00	-9.00	0.00	0.00	-9.00	-22.50	0.00	-29.00	0.00	0.00	0.00	0.00	0.00	-0.50	0.00
Implicit Setup	-1.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Implicit Teardown	-3.00	0.00	0.00	0.00	0.00	-3.00	0.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00	4.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00
Inline Setup	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-3.00	-3.00	2.00	-3.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Merge Test Method	3.00	0.00	0.00	0.00	0.00	0.00	-3.00	0.00	-3.00	-6.00	0.00	0.00	-3.00	0.00	-3.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Parameterized Test	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Prebuilt Fixture	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Print to Assertion	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-1.00	0.00	0.00	0.00	0.00	-1.00	0.00
Replace Dependency with Test Double	0.00	-1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-1.00	0.00	0.00
Replace annotation with assertThrow	-0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Split Parameterize Test	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	3.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Split Test Method	0.00	0.00	0.00	0.00	0.00	-3.00	1.50	0.00	0.00	0.00	0.00	7.00	10.00	2.00	10.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Suite Fixture Setup	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Test Helper	-37.00	-11.00	0.00	0.00	0.00	-2.00	0.00	0.00	-13.00	0.00	0.00	0.00	-49.00	0.00	-52.00	-3.00	0.00	0.00	0.00	0.00	-3.00	0.00
Verification Method	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.00	0.00	0.00	0.00	0.00	0.00	3.00	0.00

Test Smell

Change in Test Smell Frequency

10
0
−10
−20
−30
−40
−50